SMART KINESIO TAPE WITH MOTION SENSORS AND LED FEEDBACK FOR REAL TIME KNEE MONITORING

A PRODUCT DEVELOPMENT REPORT

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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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BACHELOR OF ENGINEERING

By

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BONAFIDE CERTIFICATE

Certified that this product development report "SMART KINESIO TAPE WITH MOTION SENSORS AND LED FEEDBACK FOR REAL TIME KNEE MONITORING" is the Bonafide work of LAASYA M who carried out the Product development work under my supervision.

HEAD OF THE DEPARTMENT	**SUPERVISOR**
Professor	Professor
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INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION BY THE CANDIDATE

The undersigned declares that the "**SMART KINESIO TAPE WITH MOTION SENSORS AND LED FEEDBACK FOR REAL TIME KNEE MONITORING**" project submitted for the SPIC319-MINI PROJECT FOR BIOMEDICAL APPLICATIONS course is our original work. We carried out this project under the guidance of **Dr ARANGASAMY R** from April to September 2025 and it has been completed.

Signature of the Candidate

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SMART KINESIO TAPE WITH MOTION SENSORS AND LED FEEDBACK FOR REAL TIME KNEE MONITORING

1. EXECUTIVE SUMMARY:

Knee joint mobility plays a vital role in walking, balance, rehabilitation, and overall functional movement. Limitations in knee flexion and extension can significantly affect daily activities, recovery after injury, and long-term joint health. This report presents a wearable smart knee monitoring device designed to track real-time knee joint motion, specifically flexion and extension, using embedded motion sensing and microcontroller-based processing.

The device is compact, lightweight, and non-restrictive, enabling continuous monitoring of knee movement during daily activities and rehabilitation exercises. Designed for ease of use, the system operates on a rechargeable power source and can be comfortably worn for extended periods without interfering with natural motion. By providing accurate joint motion data, the product enables users to observe range of motion and movement patterns outside clinical or laboratory environments.

Developed using commercially available components and a wearable form factor, the smart knee monitoring device offers a practical and accessible solution for knee motion tracking. The product supports mobility monitoring and rehabilitation progress assessment, contributing to improved movement awareness and functional recovery while promoting independent, real-world joint monitoring.

2.INTRODUCTION:

PROBLEM STATEMENT: Restricted or improper knee joint movement significantly affects mobility, balance, and functional activities such as walking, climbing stairs, and rehabilitation exercises. Accurate monitoring of knee flexion and extension is essential for assessing joint mobility and recovery progress. However, existing knee motion assessment systems are largely limited to clinical or laboratory environments, making them expensive, non-portable, and unsuitable for continuous use. General-purpose fitness wearables do not provide joint-specific motion data, leaving users without reliable real-time feedback on knee movement during daily activities. A wearable, affordable, and non-invasive solution for real-time knee joint motion monitoring is therefore required.

PURPOSE: The purpose of this product is to develop a wearable smart knee monitoring device capable of tracking real-time knee flexion and extension using embedded motion sensing and processing. The device is intended to help users monitor joint movement patterns and range of motion during daily activities and rehabilitation exercises, without dependence on clinical motion analysis systems or specialized supervision.

SCOPE: The smart knee monitoring device is designed for individuals requiring knee motion tracking for mobility monitoring, rehabilitation support, or movement assessment. Its compact and lightweight wearable design enables continuous monitoring without restricting natural knee movement. The system is suitable for use in both home and supervised settings and is developed using cost-effective, commercially available components. The product is adaptable and expandable for future enhancements, while remaining focused on non-invasive, real-time knee joint motion tracking.

3. GPCU (Gap Analysis, Product Development, Comparison, Uniqueness)

Market Gap:

Many knee joint motion assessment methods currently in use rely on clinical motion capture systems, optical tracking setups, or supervised rehabilitation equipment, which are expensive, bulky, and restricted to laboratory or hospital environments. While general fitness wearables are widely available, they do not provide knee-specific motion tracking or accurate flexion–extension data. As a result, users lack access to continuous, real-time monitoring of knee joint movement during daily activities and rehabilitation exercises. There is a clear gap for a wearable, non-invasive, and affordable knee monitoring device that enables real-time joint motion tracking outside clinical settings.

Product Description: The wearable smart knee monitoring device uses a microcontroller-based control unit integrated with a motion sensor to measure knee joint flexion and extension in real time. The sensor captures angular movement of the knee, and the processed data provides accurate joint motion information during walking, bending, and rehabilitation exercises. The device is powered by a rechargeable battery and includes a simple power control mechanism for user operation. Designed with a lightweight and ergonomic knee-mounted form factor, the system ensures comfort during prolonged use while maintaining reliable motion tracking. The product offers a cost-effective and practical solution for continuous knee joint motion monitoring.

Competitive Analysis:

Feature	**Existing Product A: Smart Knee Brace**	**Existing Product B: Motion-Capture IMU**	**Existing Product C: Fitness Band**	**Proposed Product: Smart Kinesio Tape**
Wearable Format	Bulky brace	Multi-sensor straps	Wristband	Kinesio tape with tiny module
Angle Tracking	Yes	Yes	No	Yes
Real-Time Feedback	Limited	Yes	Nil	Yes
Comfort	Medium	Low	High	High, lightweight
Images				

Uniqueness of the Product: The wearable smart knee monitoring device provides real-time knee joint motion tracking through a compact and non-intrusive wearable design. Unlike laboratory-based systems, it enables continuous monitoring in everyday environments, and unlike general fitness trackers, it focuses specifically on knee flexion and extension. The lightweight, rechargeable, and user-friendly design allows prolonged use without restricting natural movement. By combining accuracy, portability, and simplicity, the product offers an accessible solution for individuals seeking reliable knee motion monitoring for mobility assessment and rehabilitation support.

4. DESIGN AND ENGINEERING STANDARDS:

4.1. Electrical Safety & Performance

The smart knee monitoring device operates on a low-voltage rechargeable Li-ion battery integrated with a protection circuit to ensure safe operation. The power system supplies the microcontroller and motion sensor used for knee joint motion tracking. Electrical insulation, current limiting, and thermal protection mechanisms are incorporated to prevent overheating, short circuits, and electrical hazards during prolonged use.

Standard Reference:
- IEC 60601-1: Medical electrical equipment – General safety and essential performance requirements
- IEC 61010: Safety requirements for electrical equipment for measurement, control, and laboratory use

4.2. Sensor Accuracy and Integration

The device uses an inertial motion sensor (IMU) to accurately detect knee joint flexion and extension angles in real time. The microcontroller processes angular data to determine joint movement patterns and range of motion. Calibration routines are implemented to maintain accuracy and consistency during continuous operation.

Standard Reference:
- ISO/TR 17427: Data accuracy requirements for intelligent sensing systems
- IEEE 11073: Health informatics – Personal health device communication

4.3. Usability & Human Factors

The smart knee monitoring device is designed to be lightweight, flexible, and ergonomically positioned near the knee joint to ensure user comfort. The wearable form factor, including the Smart Kinesio Tape design, allows unrestricted natural movement. Minimal user controls and simple indicators ensure ease of use for individuals undergoing rehabilitation or mobility monitoring.

Standard Reference:

● IEC 62366: Usability engineering for medical devices

● ISO 9241: Ergonomics of human-system interaction

4.4. EMC and Electromagnetic Safety

The electronic circuitry is designed with shielded connections, noise filtering, and optimized PCB layout to minimize electromagnetic interference. These measures ensure reliable sensor operation and prevent emissions that could interfere with nearby electronic or medical equipment.

Standard Reference:

● IEC 61000-6-1: Electromagnetic compatibility – Immunity requirements

● IEC 60601-1-2: Electromagnetic compatibility requirements for medical electrical equipment

4.5. Risk Management and User Safety

Risk management measures include battery management circuitry (BMS), safe voltage limits, and protective housing for electronic components. Fail-safe software routines and watchdog timers are implemented to handle unexpected faults such as sensor errors or power instability, ensuring user safety during operation.

Standard Reference:

● ISO 14971: Risk management for medical devices

● UN 38.3: Lithium battery transport and safety testing

4.6. Biocompatibility and Hygiene

Materials used in the smart knee monitoring device, including the adhesive layer and wearable substrate, are selected to be hypoallergenic, breathable, and non-toxic. These materials prevent skin irritation and allow prolonged contact during daily use and rehabilitation activities.

Standard Reference:

● ISO 10993: Biological evaluation of medical devices

4.7. Software and Embedded System Reliability

The embedded firmware includes error handling, calibration routines, and watchdog mechanisms to ensure stable and reliable operation. Power management strategies are implemented to optimize battery life while maintaining continuous knee motion monitoring.

4.8. Standard Reference

- IEC 62304: Medical device software – Life cycle requirements
- ISO 13485: Quality management systems for medical devices

5. Technical Sketches and Diagrams

5.1-D MODEL

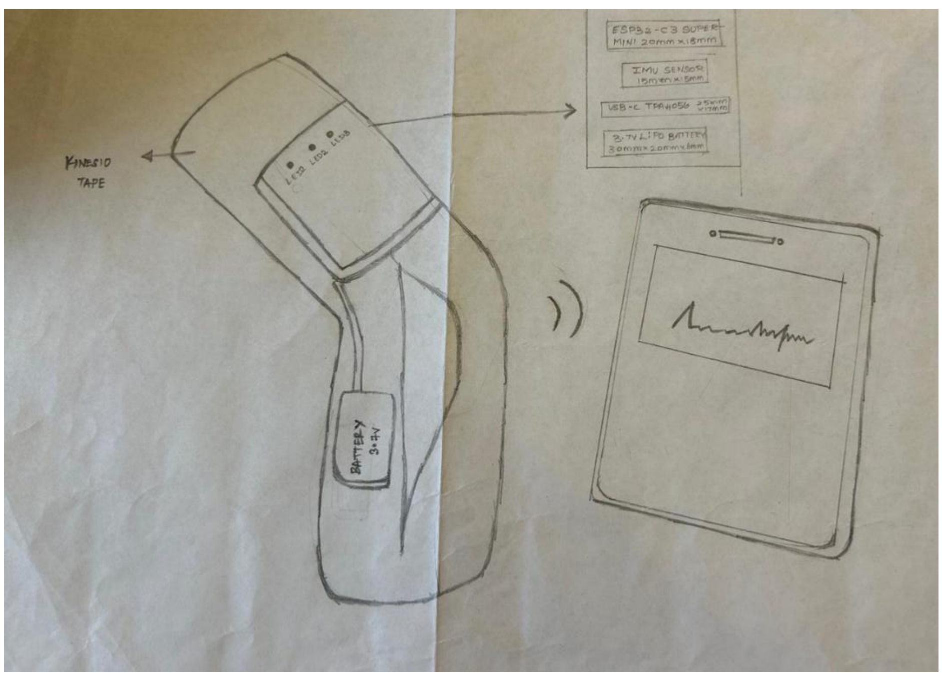


Fig 5.1: *Labelled Sketch of Smart Kinesio Tape with Motion Sensors*

5.2 3D DESIGN

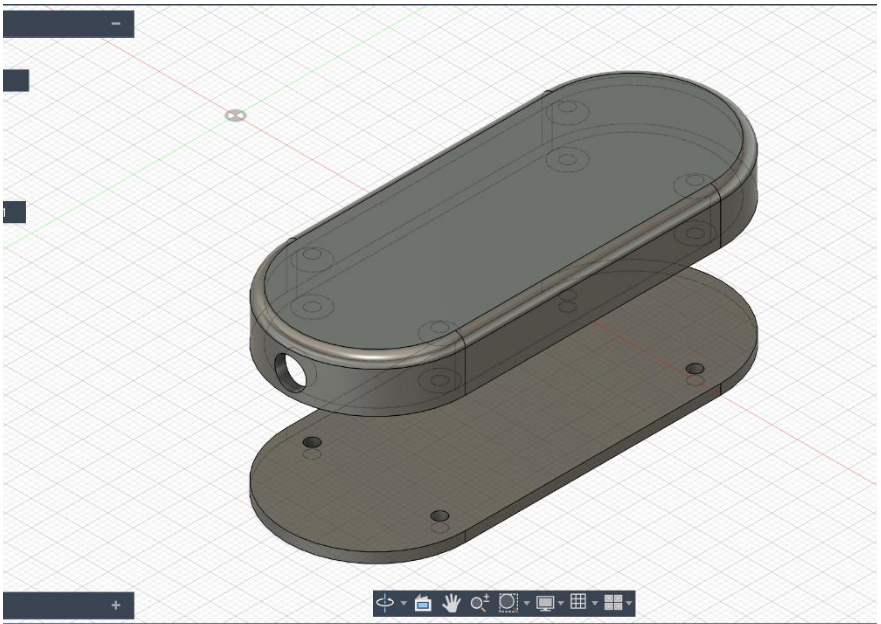


Fig 5.2: *3D Design of the MCU Module*

5.3 Product Image:

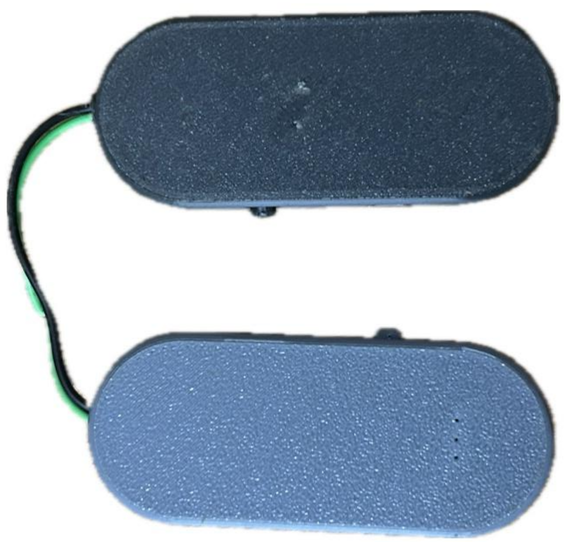


Fig 5.3.: *MCU Module*

Fig 5.3.2: *Back View of Advanced Tremor Management Gloves*

6. FUNCTIONAL PROTOTYPE:

COMPONENTS AND ITS DESCRIPTION

1. ESP32 Super-Mini C3 (Microcontroller)

Function:

Acts as the central control unit of the smart knee monitoring device.

Usage:

● Compact size suitable for wearable and embedded applications

● Low power consumption with sleep modes

● Integrated processing capability for real-time motion data handling

Role:

Collects motion data from the sensor, processes knee joint flexion and extension information, and controls overall system operation.

Fig 6.1 *ESP32 Super-Mini C3*

2. MPU6050 (Accelerometer + Gyroscope Sensor)

Function:

Detects motion and orientation of the knee joint.

Usage:

● Combines a 3-axis accelerometer and 3-axis gyroscope in a single module

● High sensitivity suitable for capturing knee flexion–extension movements

Role:

Provides real-time inertial data required to calculate knee joint angles and movement patterns.

Fig 6.2 *MPU6050*

3. TP4050 Battery Charging Module

Function:

Manages charging of the rechargeable lithium-ion battery.

Usage:

● Provides constant-current/constant-voltage (CC/CV) charging

● Protects the battery from overcharging

● Suitable for single-cell Li-ion batteries

Role:

Ensures safe and reliable charging of the battery while supporting portable operation of the device.

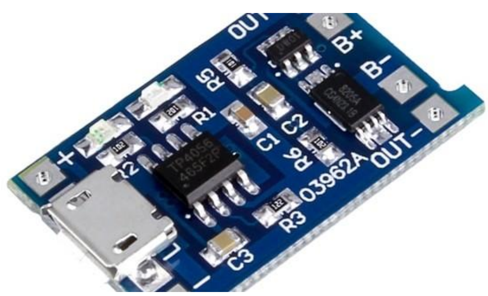


Fig 6.3 *TP4050 Charging Module*

4. Flexible Wearable Substrate (Smart Kinesio Tape / Knee Strap)

Function:

Serves as the physical interface between the device and the knee joint.

Usage:

● Lightweight, flexible, and skin-friendly

● Allows natural knee movement without restriction

Role:

Holds and positions the electronic components securely near the knee joint to enable accurate motion sensing.

Fig 6.4 *Wearable Substrate*

5. 3.7V Lithium-Ion Battery

Function:

Powers the entire smart knee monitoring system.

Usage:

● Rechargeable

● Compact and lightweight

● Suitable for wearable electronics

Role:

Supplies consistent power to the ESP32 Super-Mini C3, motion sensor, and charging module during operation.

Fig 6.5 Lithium-Ion Battery

6. On/Off Switch

Function:

Allows the user to manually turn the smart knee monitoring device ON or OFF.

Usage:

● Provides simple control for the user

● Helps save battery power when the device is not in use

Role:

Acts as a user interface element for controlling the power flow to the system, enabling safe and convenient operation of the device.

Fig 6.6 *On/Off Switch*

7. Wires and Connectors

Function:

Connect all electronic components electrically within the system.

Usage:

● Flexible and easy to solder

● Suitable for compact wearable assemblies

● Necessary for completing the circuit connections

Role:

Ensure stable electrical connections between the ESP32 Super-Mini C3, motion sensor, battery, TP4050 charging module, and On/Off switch, enabling reliable system operation.

Fig 6.7 *Wires and Connectors*

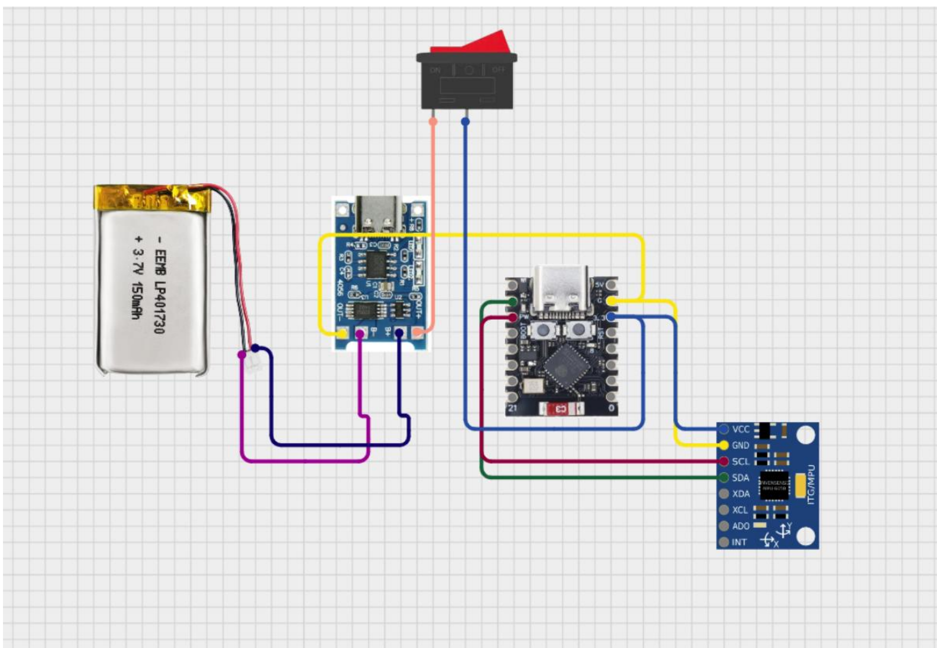
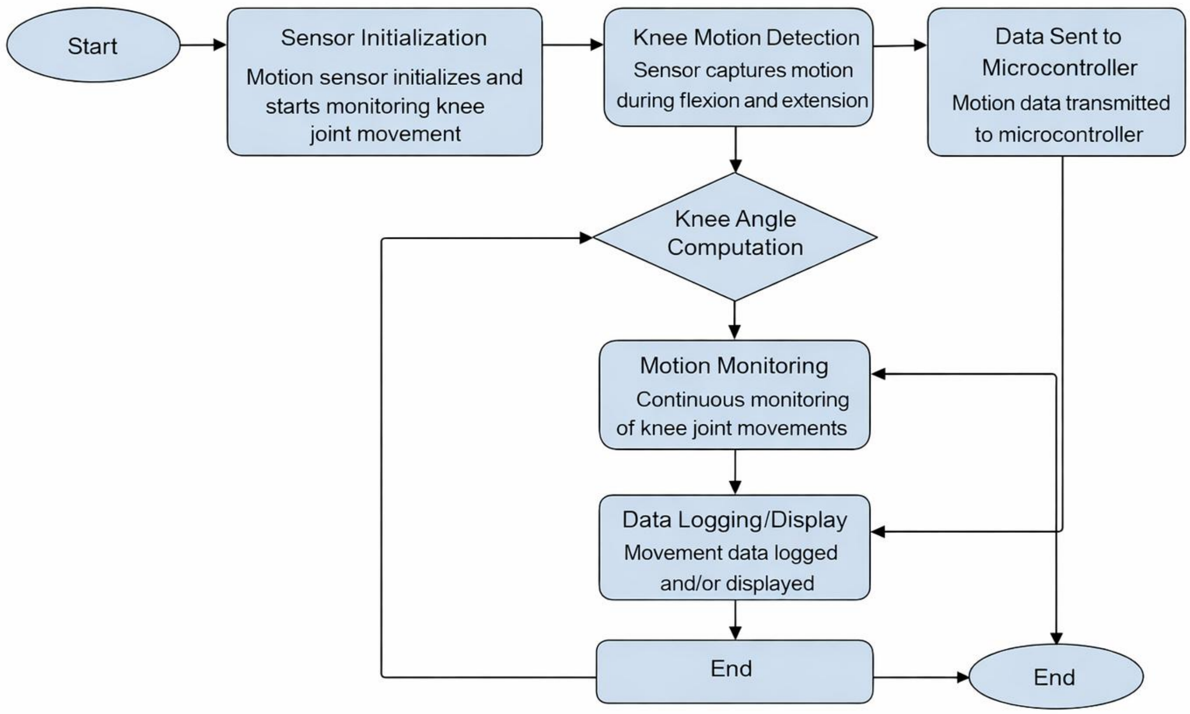


Fig 6.8: *Design Circuit of electrical*

7. PROCESSED WORK:



8. TESTING AND VALIDATION:

8.1. Knee Motion Detection Accuracy

Objective:

Evaluate the ability of the motion sensor (MPU6050) to accurately detect knee joint flexion and extension angles.

Method:

Controlled knee movements were performed through predefined angular positions. Sensor output was recorded and compared with reference angle measurements to verify accuracy and consistency of knee motion tracking.

8.2. Angle Computation and Motion Tracking Validation

Objective:

Ensure accurate computation of knee joint angles from raw sensor data processed by the ESP32 Super-Mini C3 microcontroller.

Method:

Real-time inertial data from the MPU6050 was processed to calculate flexion–extension angles. The computed angles were monitored during repeated knee movements to validate stable and continuous motion tracking.

8.3. Power Efficiency and Battery Endurance

Objective:

Validate the operational duration of the smart knee monitoring device under continuous motion sensing and data processing.

Method:

The system was powered using a fully charged 3.7V lithium-ion battery. Voltage levels and operating time were monitored during continuous use to estimate battery endurance and typical usage duration.

8.4. User Feedback and Wearability Test

Objective:

Assess comfort, usability, and practicality of the wearable device during daily activities.

Method:

Volunteers wore the smart knee monitoring device for extended durations during activities such as walking, sitting, and light exercise. Feedback was collected regarding comfort, flexibility, ease of wearing, and any restriction to natural knee movement.

9. Conclusion and Future Work

The proposed Wearable Smart Knee Monitoring Device provides a practical and non-invasive solution for continuous monitoring of knee joint motion. By integrating the ESP32 Super-Mini C3 microcontroller and MPU6050 motion sensor**,** the system accurately tracks knee flexion and extension in real time. The lightweight, wearable design using a flexible substrate ensures user comfort and allows natural movement during daily activities and rehabilitation exercises. The rechargeable power system supports prolonged use, making the device suitable for real-world knee motion monitoring outside clinical environments.

Future work may focus on enhancing motion analysis algorithms to improve accuracy and reliability across different movement patterns and activity levels. Further miniaturization of electronic components and the development of custom PCBs can reduce size and improve wearability. Integration with mobile applications for data visualization and long-term tracking can expand usability. Additional validation studies with a larger user base may be conducted to assess long-term performance, comfort, and user acceptance, supporting future commercialization of the device as an accessible and user-friendly knee motion monitoring solution.

10. References:

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9. IEC 62304 “Medical device software – Software life cycle processes.” International Electrotechnical Commission.

10. ISO 14971 “Medical devices – Application of risk management to medical devices.” International Organization for Standardization.

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(PRODUCT IMAGE)

Fig1.: *MCU Module*

Appendix: -

Coding:

- Manual

- Battery details

- Student Contact details